

# Experiment 'p\_6\_1\_a\_process\_array\_even\_indexed\_array'

## Results

June 4, 2026

**Experiment outcome:** SUCCESS  
**Bad responses:** 0  
**Responses containing assume :** 0  
**Responses containing expect :** 0  
**Resolution attempts:** 1  
**Hard fails (resolution):** 0  
**Soft fails (resolution):** 0  
**Verification attempts:** 1

## Problem Specification

**Problem name:** p\_6\_1\_a\_process\_array\_even\_indexed\_array

**Natural language statement:** Write a method that takes an array of ten integers and then returns every element at an even index.

**Method signature:** p\_6\_1\_a\_process\_array\_even\_indexed\_array(inputs: array<int>) returns (even\_indexed: array<int>)

### Ensures

- even\_indexed[..] == [inputs[0], inputs[2], inputs[4], inputs[6], inputs[8]]

### Requires

- inputs.Length == 10

## GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

### Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes an array of ten integers and then returns every element at an  
→ even index.

The signature should be:

p\_6\_1\_a\_process\_array\_even\_indexed\_array(inputs: array<int>) returns (even\_indexed: array  
→ <int>)

The method should respect the following contract:

requires inputs.Length == 10, ensures even\_indexed[..] == [inputs[0], inputs[2], inputs  
→ [4], inputs[6], inputs[8]]

Produce and show only the Dafny body of this method, including the curly braces that  
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor  
→ 'expect' anywhere in your code.

### GenAI → Program

**System fingerprint:** null

**ID:** chatcmpl-DmzzLmwaQrmivVOFE1O8Xnf2Pp1W7

**Model:** gpt-5.5-2026-04-23

**Created at:** 1780570399

**Finish reason:** stop

**Usage:** 340 tokens in, and 546 tokens out

```
{
  even_indexed := new int[5];
  even_indexed[0] := inputs[0];
  even_indexed[1] := inputs[2];
  even_indexed[2] := inputs[4];
  even_indexed[3] := inputs[6];
  even_indexed[4] := inputs[8];
}
```

### Final Program

```
method p_6_1_a_process_array_even_indexed_array(inputs: array<int>) returns (even_indexed
→ : array<int>)
  requires inputs.Length == 10
  ensures even_indexed[..] == [inputs[0], inputs[2], inputs[4], inputs[6], inputs
→ [8]]
```

```
{
  even_indexed := new int[5];
  even_indexed[0] := inputs[0];
  even_indexed[1] := inputs[2];
  even_indexed[2] := inputs[4];
  even_indexed[3] := inputs[6];
  even_indexed[4] := inputs[8];
}
```

## Total Token Usage

**Input tokens:** 340

**Output tokens:** 546

**Reasoning tokens:** 444

**Sum of ‘total tokens’:** 886

## Experiment Timings

**Overall Experiment** started at 1780570398410, ended at 1780570438196, lasting 39786ms (39.79 seconds)

**Iteration #1** started at 1780570398414, ended at 1780570438196, lasting 39782ms (39.78 seconds)